

Math Assignment

BSCS-1H

Bazil-Uddin-Lohan

246-0559.

Q1. (a). $f(x) = \sqrt{4-x}$

Domain $\leftarrow x \in [0, 4]$

Range $\leftarrow [0, 2]$

(b) $f(x) = \frac{2-3x}{7-2x}$

$7-2x \neq 0$

$7-2x$

$x \neq 7/2$

Domain $\leftarrow x \in \mathbb{R} - \{7/2\} \rightarrow (-\infty, 7/2) \cup (7/2, +\infty)$

Range $\leftarrow (-\infty, +\infty) \rightarrow (-\infty, 7/2) \cup (7/2, +\infty)$ OR $(-\infty, +\infty)$

Q2. $f(x) = \frac{4x}{3-x}$

x	y
0	0
1	2

- As at all values of domain are mapped on some values of codomain

- It also passes horizontal line test

- Every domain value of x is been mapped onto unique codomain value.

Q3. (a). Figure 1 $\rightarrow x=1, x=0, x=1$

Figure 3 $\rightarrow x=0, x=1$

Figure 2 $\rightarrow x=1, x=2, x=3$

(b) Figure 1 $\rightarrow$ at $x=-1$, The left side limit $\neq$ Right side limit and $f(-1)$ value is not equal to limit which does not exist.
at $x=0$, The ^{left} side limit is approaching to $+\infty$ and right side limit to 3 so $\lim_{x \rightarrow 0} f(x)$ DNE so $\lim_{x \rightarrow 0} f(x)$ does not exist.

At $x=1$, $\lim_{x \rightarrow 1^-} f(x) = 1$, $\lim_{x \rightarrow 1^+} f(x) = 1$ but $f(1) = 2$ which is not equal to $\lim_{x \rightarrow 1} f(x)$ so discontinuous.

Figure 2 $\rightarrow$ at $x=1$, $\lim_{x \rightarrow 1^-} f(x) = 4$, $\lim_{x \rightarrow 1^+} f(x) = 4$ but $f(1)$ value is not defined so $\lim_{x \rightarrow 1} f(x)$ does not exist.

at $x=2$, $\lim_{x \rightarrow 2^-} f(x) = 3$, $\lim_{x \rightarrow 2^+} f(x) = 3$ but $f(2) = 2$ so $\lim_{x \rightarrow 2} f(x) \neq f(2)$ so not continuous.

MAAZ

at $x=3$, $\lim_{x \rightarrow 3^-} f(x) = 0$, $\lim_{x \rightarrow 3^+} f(x) = 1$ so discontinuous as left side limit $\neq$ Right side limit so limit does not exist.

Figure 3 $\rightarrow \lim_{x \rightarrow 0^-} f(x) = -1, \lim_{x \rightarrow 0^+} f(x) = 0$ so limit not exist that makes it discontinuous.

$\lim_{x \rightarrow 1^-} f(x) = 2, \lim_{x \rightarrow 1^+} f(x) = 2$ so limit exists but $f(1) = 1$ that makes it discontinuous as $f(1)$ should be equal to $\lim_{x \rightarrow 1} f(x)$

(c) Ans: Figure 1 = $(-\infty, -1) \cup [-1, 0) \cup (0, 1) \cup (1, +\infty)$

Figure 2 = $(0, 1) \cup (1, 2) \cup (2, 3) \cup (3, 4)$

Figure 3 = $[-1, 0) \cup (0, 1) \cup (1, 2) \cup [2, 3]$

(d) Ans: $f(x) = \begin{cases} x^2 - 1 & -1 \leq x < 1 \\ 2x & 0 < x < 1 \\ 1 & x = 1 \\ -2x + 4 & 1 \leq x < 2 \\ 0 & 2 < x < 3 \end{cases}$

(4) Ans: $f(x) = \begin{cases} 4x^2 + ax + b & \text{if } x < 3 \\ a + b - 2 & \text{if } x = 3 \\ 2x^3 + bx + a & \text{if } x > 3 \end{cases}$

$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$
so $\lim_{x \rightarrow 3} f(x)$ will exist $\lim_{x \rightarrow 3} f(x) = f(3)$

$$\lim_{x \rightarrow 3^-} f(x) = 4(3)^2 + a(3) + b = 36 + 3a + b$$

$$\rightarrow 36 + 3a + b = a + b - 2$$

$$\lim_{x \rightarrow 3^+} f(x) = 2(3)^3 + b(3) + a = 54 - 3b + a$$

$$2a = -38 \rightarrow a = -19$$

$$\frac{1}{54} - 3(b) - 19 = -19 + b - 2$$

At $a = -19, b = 14$ function is continuous.

$$b = 14$$

05) (a) Ans: $\lim_{x \rightarrow +\infty} \frac{2x^4 - 4x^2 + 5}{3x^4 - 7x + 2}$ = multiply by $(1/x^4)$ $\rightarrow \frac{2 - 4/x^2 + 5/x^4}{3 - 7/x^3 + 2/x^4} = \frac{2 - 0 + 0}{3 - 0 + 0} = \frac{2}{3}$

(b) Ans: $m = \lim_{x \rightarrow 3^-} \frac{x^3 - 5x + 4}{x^3 - 8x - 3}$

$$\text{if } x = 2.9 \rightarrow \frac{(2.9)^3 - 5(2.9) + 4}{(2.9)^3 - 8(2.9) - 3} = -7.7$$

$$\text{At } x = 2.99 \rightarrow -83, \text{ At } x = 2.999 \rightarrow -841 \rightarrow \text{so } -\infty$$

MAAZ

Basil-Uddin-Uhan
246-0559, BSCS-14.

Date

$$(c). \lim_{x \rightarrow 1} f(x) = \frac{3x^3 - 7x^2 + 6x - 2}{x - 1} = 1$$

$$\text{so } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 1$$

$$\lim_{x \rightarrow 1^-} f(x) = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = 1$$

$$\text{At } x = 0.9 \rightarrow 0.83$$

$$\text{At } x = 1.1 \rightarrow 1.23$$

$$\text{At } x = 0.99 \rightarrow 0.98$$

$$x = 1.001 \rightarrow 1.002$$